



# GyaanVault

SPPU M.Sc. Data Science — Free Study Material

# Big Data

**DS-560-MJ**

Interview Preparation — Topic-wise Q&A

Semester II · Elective

Curated interview questions and answers for Big Data, organised by topic from the SPPU M.Sc. Data Science syllabus. Ideal for viva, placement and internship preparation.

*Tip: read the answer, then close the PDF and try to explain each concept aloud in your own words.*

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## Hadoop & Spark

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### **Q. What is Hadoop?**

An open-source framework for distributed storage (HDFS) and processing (MapReduce) of large datasets across clusters of commodity hardware.

### **Q. What is HDFS?**

The Hadoop Distributed File System — stores data as blocks replicated across nodes for fault tolerance and high throughput.

### **Q. How does MapReduce work?**

Map processes input into key-value pairs; Shuffle groups by key; Reduce aggregates values per key — all in parallel across nodes.

### **Q. Why is Spark faster than MapReduce?**

Spark processes data in-memory (RDDs/DataFrames) and uses a DAG execution engine, avoiding the repeated disk I/O of MapReduce.

### **Q. What is the Hadoop ecosystem?**

Components like HDFS, MapReduce, YARN, Hive (SQL), Pig, HBase (NoSQL), Mahout (ML) and Spark.

